

pd**bt**hink

Tool-free reasoning over protein structures

117 semantic instances | 20 question families | 49 protein groups

59% of instances on structures released after 2023-01

github.com/Open-Athena/pdbthink

THE FINDING

The scores are output budgets, not capabilities

GPT-6 Astra scores 0.957 across the 20 families.

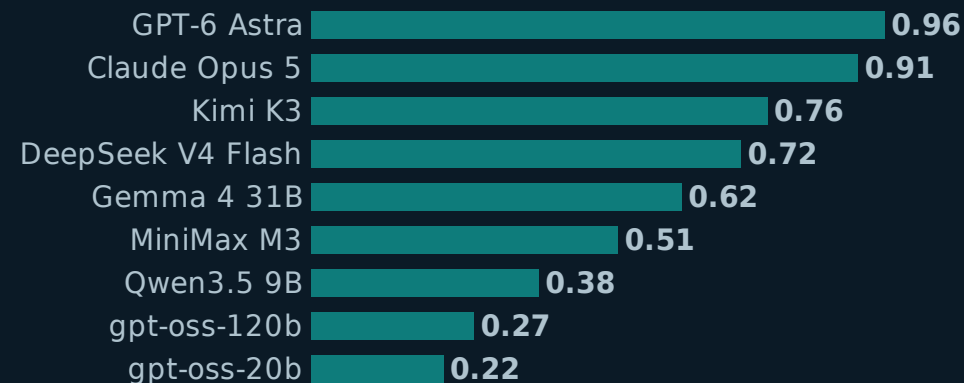
Restricted to responses that finished: 0.957.

Truncation and inability score identically — zero.

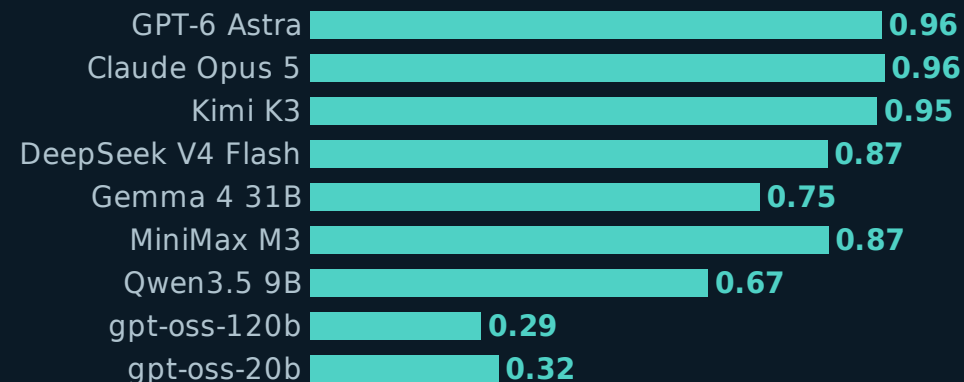
In every family where this model scores badly, the format-error count equals the truncation count. It is not failing to reason about interfaces; it never finished writing the answer.

A benchmark of reasoning models that does not report truncation separately is reporting an output budget.

AS SCORED



COMPLETED RESPONSES ONLY



THE EXPERIMENT

Give the cut-off answers more room

Every prompt below scored exactly 0.000 the first time, because the response hit the output cap before reaching a FINAL line. Re-run at a larger budget, unchanged in every other respect:

MODEL	PROMPTS	BUDGET	BEFORE	AFTER	CUT OFF IN TIER	CUT OFF OVERALL
Claude Opus 5	13	128k	0.000	0.726	1/13	1/253
Kimi K3	58	64k	0.000	0.619	13/58	15/253
	11	256k	0.000	0.685	0/11	4/253
MiniMax M3	31	128k	0.000	0.614	4/31	75/253
Qwen3.5 9B	26	128k	0.000	0.365	0/26	134/253
gpt-oss-120b	8	40k	0.000	0.083	3/8	9/253
gpt-oss-20b	86	40k	0.000	0.038	60/86	74/253

A zero from a cut-off answer is recoverable; a zero from a wrong answer is not. For the stronger models most of that zero was budget. The two right-hand columns differ because the ladder is incomplete: not every truncated prompt was re-run, so the overall count is the honest one.

The two gpt-oss rows are the control, and they are inconclusive rather than negative: their 131,072-token context sits beside an 87,500-token output, so the model had the room to finish in 25% of cases. The other models, with their 20k- to 256k-token context, hit the output cap before reaching a FINAL line.

ALL RUNS

Results

MODEL	RENDERS	AS SCORED	WITH RE-RUN	FINISHED ONLY	TRUNCATED
GPT-6 Astra	253	0.957	—	0.957	0/253
Claude Opus 5	253	0.913	0.952	0.963	13/253
Kimi K3	253	0.762	0.930	0.950	60/253
DeepSeek V4 Flash	238	0.718	—	0.867	45/238
Gemma 4 31B	156	0.618	—	0.754	19/156
MiniMax M3	253	0.512	0.594	0.868	102/253
Qwen3.5 9B	253	0.380	0.420	0.667	160/253
gpt-oss-120b	253	0.271	0.272	0.287	14/253
gpt-oss-20b	253	0.221	0.240	0.315	100/253
Marin 32B (base) (0/20 families — not comparable)	36	0.000	—	—	22/36

PER FAMILY

GPT-6 Astra: solved outright, or never finished

16 of 20 families at 0.95 or above

G01	G02	G03	G04	N01	P01	P02	P03	S01	S02	S04
S06	S07	S08	S09	T01						

the rest, and what they cost in truncation

FAMILY	SCORE	FINISHED ONLY	TRUNCATED
S05	0.625	0.625	0/8
MECH	0.828	0.828	0/18
I01	0.848	0.848	0/11
S03	0.875	0.875	0/8

Read the middle column

Every one of these families recovers once the responses that hit the cap are removed.

The apparent weakness is the 32,768-token output budget, not the geometry.

This is why truncation is its own failure category rather than folded into the score:
a zero from a cut-off answer and a zero from
a wrong one mean opposite things.

CONTROLS

What the coordinates are worth — and a trap

Seven families and the six mechanistic episodes are also asked with the coordinates removed. For the automatic families the sanitised question names no protein, so what the control measures is the guessing floor.

FAMILY	FLOOR	WITH COORDS	NAIVE GAIN	TRUNCATED	TRUE GAIN
P01	0.000	1.000	+1.000	0%	+1.000
P02	0.000	1.000	+1.000	0%	+1.000
S03	0.500	0.833	+0.333	0%	+0.333
S04	0.500	1.000	+0.500	0%	+0.500
S05	0.333	0.667	+0.333	0%	+0.333
S08	0.000	1.000	+1.000	0%	+1.000
S09	0.167	1.000	+0.833	0%	+0.833

The naive gain can invert its own sign

A context-only prompt is ~200 tokens and never truncates. A coordinate prompt runs to 87,500 tokens and truncates at up to 83%. Comparing them directly pits a finished cheap answer against a cut-off expensive one, and reports that coordinates make the model worse. Conditioned on completion, the same families show the gain the control was built to measure.

A NULL RESULT

Marin 32B: the benchmark does not fit in it

Smallest prompt containing any coordinates **7,132 tokens**

Marin 32B context window **4,096 tokens**

Not one structural question fits. All 48 renders that do are context-only controls, which by construction show no structure — so the run measures a guessing floor and nothing else.

Two things make it unevaluable, both properties of the model:

It is a base model. marin-32b-instruct does not exist. It answered P01 with the entire alphabet, and elsewhere repeated the prompt until it hit the cap — all 34 truncations.

config.json claims rope_scaling to 65k; the model card says 4,096. vLLM overrides only behind a flag whose warning is that RoPE past the derived length returns NaN — and a zero from NaN is indistinguishable from a zero from inability.

METHOD

What makes these numbers reproducible

Gold answers are recomputed, never stored

Every label comes from an oracle run on the coordinates the model was shown. A rotated variant, a cropped variant and a table variant must agree or the build fails.

The answers are not in the repository

build regenerates them byte-identically and validate checks them against committed hashes. Two independent rebuilds were byte-identical.

Responses are content-addressed, not run-addressed

Keyed on model, output budget and exact prompt text — not on render id. Adding a question costs the calls for that question. Full provider bodies are kept, so reasoning traces survive for inspection.

Truncation is a category, not a score

Reported next to every number. An earlier model moved 0.475 → 0.710 with nothing changed but its output budget.

CAVEATS

What these numbers are not

A refusal is not a wrong answer

GPT-6 Astra produced no unparseable answer in 247 renders. Its eleven zeros are refusals on context-only controls — asked a question with no coordinates supplied, it replied that the answer cannot be determined, which is correct. Its floor on those families measures declining to guess, not guessing badly.

Not curator-reviewed

The 117-instance set is proposed, not accepted. A curator interface exists and decisions are being recorded; most of the set is pending.

One completion per prompt

The protocol calls for three, and ten on a reliability subset. Intervals here cover benchmark composition only, not model stochasticity.

Three runs are incomplete

A credit limit cut them off mid-batch. They cover different family subsets and are excluded from comparison rather than presented.

Budgets differ by model

Each model got the largest output budget its context allows beside an 87,500-token prompt — gpt-oss 32,768 against Kimi's 65,536. Compare truncation counts, not only scores.

S02 is thin, and narrow

Two instances against a target of four. Four of its six candidate proteins are on the burned list — their answers reached a public commit before gold was split out. Underneath that, only 8 of 181 cached structures carry a phosphorylated residue and every one is a phosphotyrosine in an SH2 domain, so the question offers three options